COMP5424 Extended Reality

Requirements and Prototype Proposal

Scenario D: Campus Emergency Preparedness

Group: Group 7
Virtual client: Harbourview University Safety and Resilience Office
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1 Client Problem and Project Scope

1.1 Client context and problem

Harbourview University Safety and Resilience Office supports students, staff, visitors, and campus spaces. Its proposed pilot concerns a fictional four-storey academic building. Current preparation relies on online information, signage, orientation material, and occasional drills; however, engagement and building familiarity vary. Users may know general guidance but still struggle to interpret environmental cues or recover when a familiar route changes. The client therefore needs a focused, low-risk practice experience rather than a complete emergency system.

1.2 Why XR is appropriate

XR enables users to practise spatial decision-making in a supervised fictional environment without revealing real campus information. Unlike a static module, it requires users to observe signs and environmental cues, select a route, and reconsider that decision after a route change. The experience complements, rather than replaces, official guidance and drills by providing a repeatable opportunity to practise and receive an explanation of the relevant cues.

1.3 Selected emergency scenario

Selected scenario: Fire evacuation

Fire evacuation was selected under Virtual Client Response A1. It provides a clear context in which evacuation signs and environmental cues can guide a route decision without relying on invented procedures, AI guidance, or dramatic effects. The learning objective is for users to interpret these cues, recognise that an initial route is unavailable, select an alternative direction, and understand the rationale through feedback.

1.4 Prototype scope and boundaries

Proposed scope:

- Primary users: international students unfamiliar with the fictional building;
- starting point: one fictional classroom and adjacent corridor;
- core decision: interpreting signs and environmental cues to select a route;
- one unavailable-route event, shown by a closed fictional route and clear signage;
- one alternative route indicated by nearby signs and visible spatial cues; and
- one fictional safe endpoint followed by an explanatory debrief.

Out of scope:

- Real campus maps or security-sensitive information;
- Official emergency certification or formal assessment;
- A complete four-storey building;
- Dynamic AI co-navigation or multiplayer interaction;
- Graphic harm, intense smoke, startling effects, or flashing stimuli.

2 User Needs and Requirements Analysis

2.1 Primary user profile

Primary user: International students unfamiliar with the fictional building.

This group represents users who may have limited familiarity with local emergency terminology, building layouts, or XR interaction. The first prototype prioritises plain English, signs combining text and symbols, captions or text feedback, high-contrast information, teleport locomotion, and pause, exit, and restart controls. These choices also support users with hearing, language, anxiety, or previous-XR-experience differences. The prototype does not claim to meet every accessibility need: complete multilingual support, personalised assistive technology, and procedures for differing evacuation requirements remain outside scope until confirmed by the client.

2.2 Stakeholder analysis

Stakeholders require guidance consistent with approved information, a calm and inclusive experience, plain language, and a short supervised session that can be launched and reset without personal accounts. The complete stakeholder analysis is provided in Appendix A.

2.3 Prioritised requirements

Table 1: Prioritised requirements

ID	Requirement	Priority	Source	Verification
R1	The prototype shall support one complete fire-evacuation practice journey.	Must	Virtual Client A1/A3	Task completion
R2	Users shall interpret evacuation signs and environmental cues.	Must	Virtual Client A1/A6	Observation and interview
R3	Users shall respond appropriately when the initial route becomes unavailable.	Must	Virtual Client A1/A6	Route decision evidence
R4	Feedback shall explain why an action or route is appropriate.	Must	Virtual Client A5	User explanation
R5	The experience shall remain calm, inclusive, and low risk.	Must	Virtual Client A2/A4	Comfort and accessibility evidence
R6	Users shall be able to pause, exit, and restart safely.	Should	Scenario description	Functional test

R1–R5 are *Must* requirements because together they create the minimum complete and testable practice journey requested by the client: users identify the situation, interpret cues, recover from a route change, receive a rationale, and remain safe and comfortable. R6 is a *Should* requirement because it supports low-risk participation and can be implemented after the core journey is stable. Features that do not directly support this journey are deferred.

3 Assumptions, Client Questions and Responses

The team asked three questions that resolve the decisions most likely to affect scope, safety, and later evaluation. Table 2 shows how each response changed the proposal rather than merely recording it. The full response-to-design mapping is in Appendix B.

Table 2: Client questions, responses, and design consequences

ID	Question	Client response	Decision taken
CQ1	Which emergency context should drive the first complete journey?	Fire evacuation or a non-fire incident requiring evacuation is permitted.	Select fire evacuation; limit the MVP to one route change in a fictional building.
CQ2	What intensity and participation model are acceptable?	The experience must be calm, inclusive, low risk, and primarily practice rather than certification.	Use a gentle alert, no smoke, flashing, harm, scores, or rankings; provide pause, exit, and restart.
CQ3	When should feedback be given and what should later evaluation examine?	Feedback should explain rationale; evaluation may examine cues, route changes, understanding, comfort, and inclusion.	Offer an optional hint after sustained hesitation and an explanatory debrief; evaluate behaviour, explanation, and comfort.

Assumptions and limitations: The building, routes, and endpoint are fictional. Users are assumed to be unfamiliar with the building and use course-supported XR equipment under supervision. The MVP contains one route change only. It does not reproduce real procedures, provide formal training, or meet every accessibility or multilingual need; these areas require future client confirmation.

4 Proposed Prototype and Core User Journey

4.1 Prototype concept

This prototype is a calm VR practice experience for international students unfamiliar with a fictional academic building. The user begins in a classroom, receives a gentle fire-evacuation alert, and leaves into an adjacent corridor. They interpret evacuation signs and visible environmental cues to choose an initial direction. That route then becomes unavailable through a closed fictional area and a clear “Route unavailable” sign. The user re-examines nearby signs, selects an

alternative direction, reaches a fictional safe endpoint, and receives an explanatory debrief. The experience remains a practice activity and does not teach unconfirmed emergency procedures.

4.2 Core user journey

Briefing → Recognise alert → Interpret cues → Choose route → Route unavailable → Find alternative → Safe endpoint → Debrief

Figure 1: Proposed core user journey

Journey details: After a short orientation, a gentle audio-visual alert starts the practice. The user leaves the classroom, interprets signs, chooses an initial direction, encounters a clearly unavailable route, and re-examines nearby cues before following an alternative direction to the fictional safe endpoint. The debrief explains the useful cues and reason for reconsideration.

4.3 Interaction and feedback design

- **Locomotion:** Teleport locomotion is the default to reduce motion discomfort and implementation risk.
- **Route interaction:** Users observe signs and cues, then use controller pointing and confirmation to move between locations.
- **Unavailable route:** A closed fictional route and a clear text-and-symbol sign indicate that the route is unavailable.
- **Feedback:** Limited prompts appear after an error or sustained hesitation; the final debrief explains relevant cues rather than marking the user correct or incorrect.
- **Accessibility:** The prototype uses clear text, captions or text feedback, high contrast, a gentle audio alert, and pause, exit, and restart controls.

5 Technical Selection and Feasibility Analysis

The prototype uses Unity LTS and course-supported XR hardware, teleport locomotion, controller selection, fixed triggers, and a simple state machine. These choices minimise motion discomfort and implementation risk while supporting the complete route-change journey. The detailed technical rationale is in Appendix C.

5.1 Feasibility analysis

Technical feasibility:

The selected scope is technically achievable within the semester. A small Unity scene, fixed teleport locations, controller selection, trigger zones, and a simple state machine can control the alert, initial route, route-unavailable state, alternative route, completion, and reset. This avoids high-risk features such as procedural navigation, AI companions, multiplayer networking, or dynamic hazards.

Time and resource feasibility:

The prototype will use course-supported XR equipment and a shared Unity project. The team will prioritise a stable end-to-end journey before visual refinement or optional language support. Reusing simple modular assets and a fixed route limits asset-production and debugging time. Additional floors, multiple route changes, and unconfirmed emergency interactions will be deferred rather than risking an incomplete build.

Operational and user feasibility:

The experience is designed for a supervised teaching or evaluation setting. A facilitator launches the session, provides a short orientation, and can reset the scene after each participant. Users can pause, exit, or restart without penalty. No account, personal identifier, or long-term performance record is required; only in-session observations and consent-based evaluation notes will be used.

5.2 Risks and mitigation

- **Scope growth:** freeze one journey and one route change; defer extra features.
- **Missed cues:** pilot-test text-plus-symbol signs and adjust visibility.
- **Discomfort:** use teleport, pause, exit, and restart controls.

The complete risk register, including technical stability and feedback clarity, is in Appendix D.

6 Evaluation and Project Plan

6.1 Evaluation focus

Table 3: Core evaluation questions and evidence

ID	Evaluation question	Evidence
E1	Can users interpret signs and cues to choose a route?	Route choices, hesitation, and participant explanation
E2	When the initial route is unavailable, can users find an alternative without excessive help?	Recognition of change and alternative route choice
E3	Is the experience clear, comfortable, and appropriately calm?	Observation and short semi-structured interview

6.2 Traceability

Each prioritised requirement maps to a prototype feature and qualitative evidence: task completion, route choices, participant explanations, and comfort feedback. The full requirement-to-feature-to-evaluation matrix is in Appendix E.

6.3 Project plan and responsibilities

Table 4: Project plan and accountable members

Period	Planned activity, accountable owner, and milestone
Weeks 1–2	**Requirements and scope** — Product Group: WENG Zhirui and SONG Zheyi confirm the client context, target user, scope, and prioritised requirements.
Weeks 2–4	**Product system design and handoff** — Product Group defines the system design, feature requirements, and development standards; Development Group confirms technical feasibility and Git workflow.
Weeks 4–6	**Core implementation and test preparation** — Development Group (SUN Jiashi and FENG Kedi) implements version-controlled features. SUN reviews every feature change before Test Group (CHU Silu and ZHU Yu) testing.
Week 7	**Quick Prototype** — Development Group integrates the reviewed build; Test Group performs acceptance; Product Group converts findings into the internal issue list for the Week 7 submission.
Weeks 8–10	**Evaluation and refinement loop** — Test Group leads evaluation and reports evidence to Product Group; Product Group refines feature requirements; Development Group implements reviewed changes in Git.
Weeks 10–12	**Improvement and final integration** — Product Group confirms final design details; Development Group implements and SUN reviews prioritised fixes; Test Group regression-tests and accepts the **Week 12 Evaluation and Technical Improvement** outcome.

Team workflow and review: The Product Group (WENG Zhirui and SONG Zheyi) designs the product system and defines feature requirements and development standards. The Development Group (SUN Jiashi and FENG Kedi) implements features and manages Git version control. The Test Group (CHU Silu and ZHU Yu) designs tests, conducts testing, and performs product acceptance. SUN Jiashi reviews every feature change before it is handed to the Test Group.

Product Group: design and feature requirements → Development Group: implement and manage in Git → SUN review → Test Group: test and accept
Test feedback → Product Group: refine design → Development Group

Figure 2: Iterative team workflow under Git version control

The Week 7 Quick Prototype is the first integration gate. Evaluation evidence from Weeks 8–10 creates the prioritized improvement backlog for the Week 12 submission; new features must support a prioritized requirement and not endanger the MVP.

A Stakeholder Analysis and Complete Requirements Matrix

Table 5: Stakeholder concerns and design implications

Stakeholder	Concern and implication
Safety and Resilience Office	Guidance and feedback must support safe decisions without presenting unconfirmed procedures as authority.
Student Services	The experience should be calm, supportive, and free from fear-based motivation.
Inclusion Officer	Information should be available through clear visual and textual alternatives, with comfort controls.
Communications Officer	Plain language, familiar symbols, and concise instructions are required.
IT and Facilities	The experience should be short, supervised, easy to launch, and easy to reset without personal accounts.

Table 6: Complete prioritised requirements matrix

ID	Requirement	Priority	Source	Verification
R1	The prototype shall support one complete fire-evacuation practice journey.	Must	A1/A3	Task completion
R2	Users shall interpret evacuation signs and environmental cues.	Must	A1/A6	Observation and interview
R3	Users shall respond appropriately when the initial route becomes unavailable.	Must	A1/A6	Route decision evidence
R4	Feedback shall explain why an action or route is appropriate.	Must	A5	User explanation
R5	The experience shall remain calm, inclusive, and low risk.	Must	A2/A4	Comfort and accessibility evidence
R6	Users shall be able to pause, exit, and restart safely.	Should	Scenario description	Functional test

B Client Responses, Assumptions, and Constraints

Table 7: Virtual-client responses and design implications

ID	Client response	Proposal implication
A1	Teams may select fire evacuation or non-fire accident evacuation.	Fire evacuation and one focused route-change journey were selected.
A2	The prototype should remain calm and avoid distress, smoke, startling effects, and flashing.	The design uses a gentle alert and clear visual cues, without smoke, flashing, or jump scares.
A3	The first prototype should primarily be practice rather than formal assessment.	Feedback is supportive; the prototype makes no certification or performance claim.
A4	Teams may select one primary user group and must state assumptions and limitations.	The target group is international students unfamiliar with the building; exclusions are explicit.
A5	Feedback should explain why an action or route is appropriate.	The prototype uses limited contextual prompts and an explanatory debrief.
A6	Evaluation may examine cues, route decisions, route changes, rationale, comfort, and inclusion.	Observed behaviour and short interviews provide qualitative evidence.

Table 8: Assumptions, constraints, and limitations

Category	Content
Confirmed information	The environment is fictional; the experience is supervised, low risk, and a practice activity; fire evacuation is permitted under A1.
Team assumption	Selected users are unfamiliar with the fictional building and can use course-supported XR equipment with staff supervision.
Design decision	The experience includes one classroom, one route change, one alternative route, and a fictional safe endpoint.
Project constraint	Semester time, course-supported equipment, team skills, and the need for a stable testable prototype limit the scope.
Limitation	The prototype does not reproduce real maps or procedures, provide formal training, or cover every accessibility and evacuation need.

C Technical Decisions and Rationale

Table 9: Technical decisions and rationale

Technical decision	Rationale	Requirement supported
Development platform	Unity 6 LTS and the course-supported XR platform(Meta Quest 3).	R1, R5
Locomotion	Teleport is the default to reduce motion discomfort and implementation risk.	R5, R6
Scene structure	One classroom, one corridor, one unavailable route, and one safe endpoint.	R1, R3
Route logic	Fixed route, triggers, and a simple state machine.	R1, R3
Interaction	Controller pointing and selection.	R2, R3
Feedback	Visual and text feedback with optional audio support.	R4, R5

D Detailed Risk Register

Table 10: Main risks and mitigation strategies

Risk	Impact	Mitigation or contingency
Scope becomes too large	Incomplete or unstable build	Freeze the MVP as one journey with one route change; defer non-essential features.
Users miss route cues	Users cannot complete the task	Test sign visibility early; use text-plus-symbol signs and adjust placement or contrast.
Motion discomfort	Reduced participation	Use teleport locomotion and offer pause, exit, and restart controls at all times.
Technical instability	Unreliable evaluation	Use fixed triggers and simple state logic; test the complete reset cycle before evaluation.
Feedback is unclear	Users do not understand the rationale	Pilot-test concise prompts and revise the final debrief using participant explanations.

E Traceability and Detailed Evaluation Plan

Table 11: Requirement-to-feature-to-evaluation traceability

Req.	Prototype feature	Evaluation question	Evidence
R1	Complete core journey	E1	Task completion
R2	Evacuation signs and environmental cues	E1	Route decisions and observed behaviour
R3	Unavailable route and alternative route	E2	Error recovery and route choice
R4	Contextual feedback and debrief	E2	Participant explanation
R5	Low-intensity and accessible design	E3	Comfort and accessibility feedback

F Detailed User Journey Flowchart

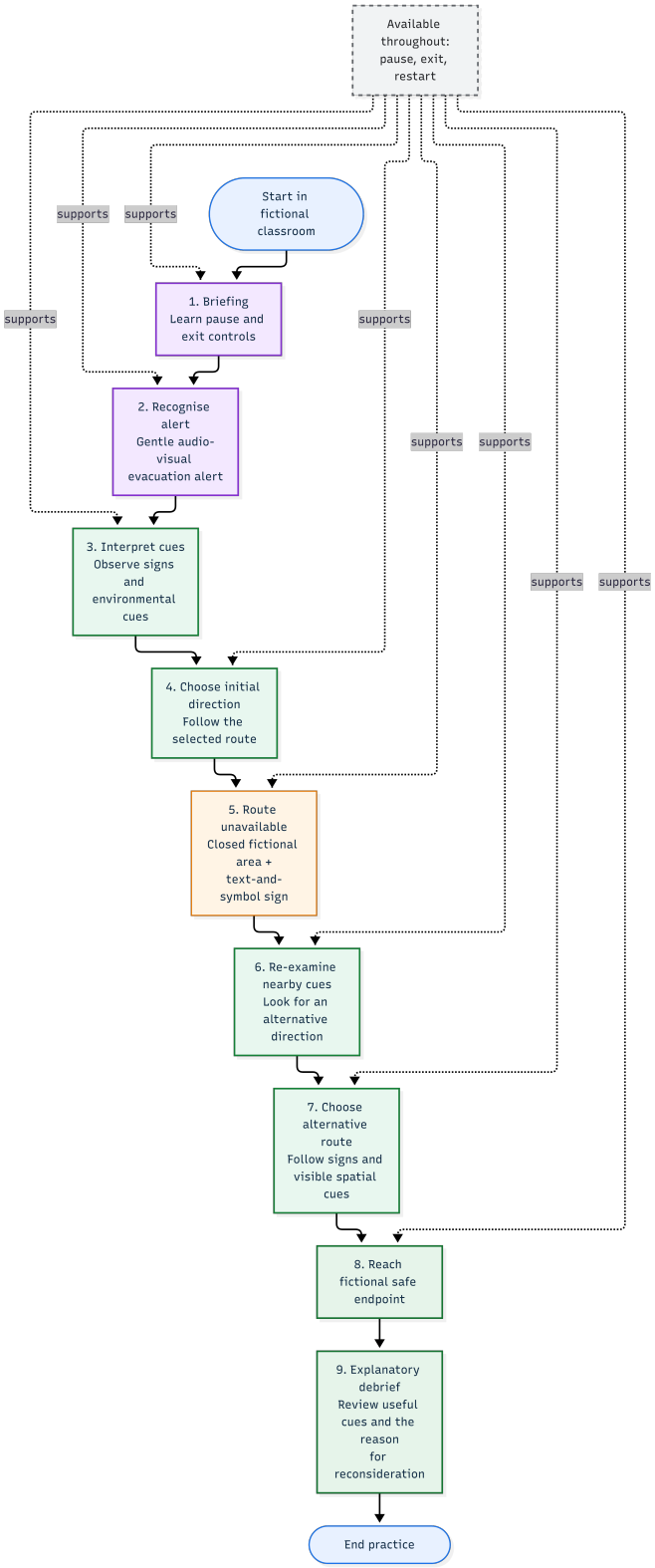


Figure 3: User journey for the fire-evacuation practice experience

References

[1] COMP5424 Group Project Description, 2026–2027.
[2] Scenario Description D: Campus Emergency Preparedness, 2026–2027.
[3] Virtual Client Responses for Scenario D, 2026.